ISLAMIC UNIVERSITY OF TECHNOLOGY

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LINEAR ALGEBRA ASSIGNMENT

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1. Background

Delivery services play a crucial role in the modern economy, with courier companies striving to efficiently transport packages from one location to another. Optimizing delivery routes is essential to reduce costs, save time, and enhance customer satisfaction. This assignment focuses on the mathematical modeling of delivery routes, taking into account factors such as distance, traffic conditions, and fuel costs.

2. Problem Statement

The challenge is to develop a mathematical framework for calculating the total cost associated with delivering packages between different locations. The key factors influencing the cost include the distances between delivery points and the fluctuating fuel prices in different areas. We will represent these factors using a distance matrix (D), where each entry dij corresponds to the distance between location i and location j, and a fuel price vector (P), where each entry pi represents the fuel price at location i.

The primary objective is to create a cost matrix (C) by multiplying the distance matrix (D) with the fuel price vector (P). Each entry cij in the cost matrix represents the cost of delivering a package from location i to location j, considering the specific distance and fuel price associated with that route.

Furthermore, we extend the scope to incorporate route planning into the framework. By utilizing the cost matrix (C), we aim to identify the most cost-effective route that visits all the delivery points. This involves optimizing the sequence of locations to minimize the total cost, contributing to an efficient and economical delivery route plan.

3. Objectives

The primary objectives of this assignment include:

- Develop a mathematical representation of delivery costs using a distance matrix (D) and fuel price vector (P).

- Calculate the total cost matrix (C) by combining distance and fuel prices for each delivery point-to-point route.

- Implement route planning strategies to identify the most cost-effective sequence of locations to visit.

- Analyze and discuss the implications of distance, fuel prices, and route optimization on the overall cost of the delivery route.

4.Data Representation

Distance Matrix (D)

The foundation of our delivery cost calculation lies in representing the distances between locations. This is achieved through the use of a distance matrix (D), where each entry dij corresponds to the distance from location i to location j.

D =

[[0, 250, 400, 180],

[250, 0, 600, 320],

[400, 600, 0, 450],

[180, 320, 450, 0]]

Fuel Price Vector (P)

Another critical factor influencing the total cost of the delivery is the price of fuel, which varies from location to location. We represent these fuel prices using a vector (P), where each entry pi denotes the fuel price in location i.

P = [1.4, 1.7, 2.0, 1.2]

Cost Matrix (C) Calculation

To derive the total cost of delivering packages between locations, we combine the information from the distance matrix (D) and the fuel price vector (P) to create a cost matrix (C). Each entry cij in the cost matrix represents the cost of delivering a package from location i to location j, considering both the specific distance and fuel price associated with that route.

C =

[[0, 425, 800, 216],

[357, 0, 1020, 384],

[560, 1020, 0, 540],

[216, 384, 540, 0]]

Interpretation and Significance

Understanding the implications of the distance matrix, fuel price vector, and cost matrix is crucial for effective route planning. This section aims to bridge the theoretical aspects of data representation with practical insights, preparing the groundwork for the subsequent chapters on cost optimization and route planning.

5.Solution

Optimal Route Planning

After representing the distances between locations using the distance matrix (D) and incorporating fuel prices with the gas price vector (P), our objective is to find the optimal route for our courier delivery service.

The task is to determine the most cost-effective route starting from the warehouse, visiting all other locations exactly once, and returning to the warehouse. This scenario aligns with the Traveling Salesman Problem (TSP).

Approach

To address the TSP, we employ the Nearest Neighbor algorithm as a simple and intuitive heuristic. This algorithm starts from an initial location, selecting the nearest unvisited location at each step until all locations are visited, forming a cycle that begins and ends at the initial location.

Application

Applying the Nearest Neighbor algorithm to our delivery scenario, commencing from the warehouse:

Step 1:

-Start at the Warehouse (City 1)

- Current location: Warehouse

- Nearest unvisited location: City 4 (Cost = 216)

- Move to City 4

Step 2:

-Visit the next nearest location from City 4

- Current location: City 4

- Nearest unvisited location: City 2 (Cost = 384)

- Move to City 2

Step 3:

- Visit the remaining location

- Current location: City 2

- Nearest unvisited location: City 1 (Cost = 357)

- Move to City 1

The resulting route is: Warehouse → City 4 → City 2 → City 1 → Warehouse. The total cost of this route, calculated using the corrected cost matrix (C), is 957.

6.Discussion

While the Nearest Neighbor algorithm provides a quick and reasonable solution, it may not guarantee the globally optimal route, as it makes local decisions at each step. Advanced algorithms, such as the branch and bound method or genetic

algorithms, could be explored for more accurate results, particularly with larger datasets.

The choice of the initial location significantly influences the outcome. Starting from the warehouse in our case, the algorithm produced a route with a total cost of 957. Evaluating alternative starting points and comparing their respective results would provide a more comprehensive understanding of the solution space.

7.Conclusion

In this section, the Nearest Neighbor algorithm was applied to determine an optimal route for our courier delivery scenario, considering both distances and fuel prices. The solution, originating from the warehouse and visiting all other locations exactly once, resulted in a total cost of 957. While this approach offers a practical delivery route plan, it's crucial to acknowledge the algorithm's limitations and explore advanced methods for further optimization.